

Aluffi Ring 1st Chapter Section 6 Exercises (half)

Zih-Yu Hsieh

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1 6.8

Problem 1

Let R be a ring. If A is any set, prove that $\text{Hom}_{R\text{-Mod}}(R^{\oplus A}, R)$ satisfies the universal property of the product R^A . So, in general $\text{Hom}_{R\text{-Mod}}(R^{\oplus A}, R) \not\cong R^{\oplus A}$.

Solution: First, let's define a projection: Given any $f \in \text{Hom}_{R\text{-Mod}}(R^{\oplus A}, R)$, define the projection $p_a : R^{\oplus A} \rightarrow R$ by $p_a(f) = f(e_a)$ for all $a \in A$. this is clearly a module homomorphism.

Now, given any $M \in R\text{-Mod}$, and for any $a \in A$ it associates with a module homomorphism $\varphi_a : M \rightarrow R$. Suppose $h : M \rightarrow \text{Hom}_{R\text{-Mod}}(R^{\oplus A}, R)$ is a module homomorphism satisfying $p_a \circ h = \varphi_a$ for all $a \in A$, then for any $m \in M$, one has the following:

$$\varphi_a(m) = p_a \circ h(m) = h(m)(e_a) \quad (1.1)$$

Which, $h(m) : R^{\oplus A} \rightarrow R$ satisfies $h(m)(e_a) = \varphi_a(m)$ for all $a \in A$. This shows that „if $h(m)$ is well-defined“, it is already uniquely determined (since it fixes all the basis elements on $R^{\oplus A}$). As a result, h is in fact uniquely determined (since it's determined for all $m \in M$).

First question: Is $h(m)$ well-defined? Yes, because the maps only tells us how it behaves on the basis elements, but didn't talk about how these elements interact. Therefore, one can define $h(m)(\sum r_a e_a) := \sum r_a h(m)(e_a) = \sum r_a \varphi_a(m)$, which is clearly well-defined as it is a finite sum. This also shows that $h(m)$ is a module homomorphism.

Second question: Is h itself a module homomorphism? Yes, because one has the following:

$$h(rm + sn)(e_a) = \varphi_a(rm + sn) = r \cdot \varphi_a(m) + s \cdot \varphi_a(n) = r \cdot h(m)(e_a) + s \cdot h(n)(e_a) \quad (1.2)$$

Hence, since it works on all the basis elements, one can say that $h(rm + sn) = r \cdot h(m) + s \cdot h(n)$ as module homomorphisms. So, h is itself a module homomorphism.

Finally, this shows that $\text{Hom}_{R\text{-Mod}}((R^{\oplus A}), R)$ is a product R^A , since it satisfies the following commutative diagram:

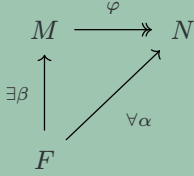
$$\begin{array}{ccccc} & & M & & \\ & \swarrow \varphi_a & \vdots \exists! h & \searrow \varphi_b & \\ R_a & \xleftarrow{p_a} & \text{Hom}_{R\text{-Mod}}(R^{\oplus A}, R) & \xrightarrow{p_b} & R_b \end{array}$$

where the dashed line is the unique h defined, and it's satisfied for all $a, b \in A$.

2 6.9

Problem 2

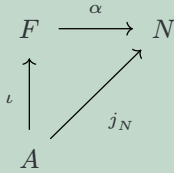
Let R be a ring, F a nonzero free R -module, and let $\varphi : M \rightarrow N$ be a homomorphism of R -modules. Prove that φ is surjective iff for all R -module homomorphism $\alpha : F \rightarrow N$, there exists an R -module homomorphism $\beta : F \rightarrow M$ such that $\alpha = \varphi \circ \beta$, i.e. F is projective.



Solution: Here, we'll let A denote the index set of the canonical basis of F .

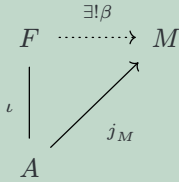
$\Rightarrow$:

First, suppose φ is surjective, then given any $\alpha : F \rightarrow N$, for all $a \in A$, since the set map $j_N : A \rightarrow N$ by $j_N(a) := \alpha(e_a)$ generates α based on the universal property of free modules as follow:



where $\iota : A \rightarrow F$ is the canonical inclusion by $a \mapsto e_a$.

Since φ is surjective, for all $a \in A$, there exists some $m_a \in M$ such that $\varphi(m_a) = \alpha(e_a)$. Which, define the set map $j_M : A \rightarrow M$ by $j_M(a) := m_a$. Based on the universal property of free modules, there exists a unique module homomorphism $\beta : F \rightarrow M$, such that the following diagram commutes:



Hence, for all $a \in A$, one has $\beta(e_a) = j_M(a) = m_a$. Hence, the composition $\varphi \circ \beta(e_a) = \varphi(m_a) = \alpha(e_a)$. Because $\varphi \circ \beta$ and α are module homomorphisms that agree on the basis elements of F , one has $\alpha = \varphi \circ \beta$.

$\Leftarrow$:

Now, for the converse we'll prove the contrapositive. Suppose φ is not surjective, there exists $n \in N \setminus \text{im}(\varphi)$. Then, take the module homomorphism $\alpha : F \rightarrow N$ by $\alpha(e_a) = n$ for all $a \in A$. For α , there doesn't exist any module homomorphism $\beta : F \rightarrow M$ such that $\varphi \circ \beta = \alpha$, since $\text{im}(\varphi \circ \beta) \subseteq \text{im}(\varphi)$, which doesn't contain $n \in \text{im}(\alpha)$.

Hence, if $\varphi : M \rightarrow N$ is not surjective, there exists module homomorphism $\alpha : F \rightarrow N$, such that for all $\beta : F \rightarrow M$, $\alpha \neq \varphi \circ \beta$. Turning the contrapositive, one gets for any module homomorphism $\alpha : F \rightarrow N$, there exists $\beta : F \rightarrow M$, such that $\alpha = \varphi \circ \beta$, implies φ is surjective.

This finishes the equivalence.

3 6.10

Problem 3

Let $M, N, Z \in \mathbf{R}\text{-Mod}$, and let $\mu : M \rightarrow Z, \nu : N \rightarrow Z$ be module homomorphisms.

Prove that $\mathbf{R}\text{-Mod}$ has *fiber product*, namely there exists an R -module $M \times_Z N$ with homomorphisms $\pi_M : M \times_Z N \rightarrow M, \pi_N : M \times_Z N \rightarrow N$, such that $\mu \circ \pi_M = \nu \circ \pi_N$; and, such pair is final with respect to such property. Namely, the following diagram commutes:

$$\begin{array}{ccccc}
 P & & & & \\
 \swarrow \exists! h & \searrow p_N & & & \\
 & M \times_Z N & \xrightarrow{\pi_N} & N & \\
 \downarrow p_M & \downarrow \pi_M & & \downarrow \nu & \\
 & M & \xrightarrow{\mu} & Z & \\
 & & & & \downarrow \nu
 \end{array}$$

Solution: First, take the product module $M \times N$ with canonical projections $\pi_M : M \times N \rightarrow M$ and $\pi_N : M \times N \rightarrow N$. Then, the composition $\mu \circ \pi_M, \nu \circ \pi_N : M \times N \rightarrow Z$ are two module homomorphisms.

Now, given the map $\alpha := \mu \circ \pi_M - \nu \circ \pi_N$, define $M \times_Z N := \ker(\alpha) \subseteq M \times N$, and take $\pi_M : M \times_Z N \rightarrow M$ and $\pi_N : M \times_Z N \rightarrow N$ as restrictions of the original projections. Then, for any $(m, n) \in M \times_Z N = \ker(\alpha)$, one has the following:

$$0 = \alpha(m, n) = \mu \circ \pi_M(m, n) - \nu \circ \pi_N(m, n) = \mu(m) - \nu(n) \quad (3.1)$$

This implies $\mu(m) = \nu(n)$ for all $(m, n) \in M \times_Z N$, hence $\mu \circ \pi_M(m, n) = \nu \circ \pi_N(m, n)$, showing the commutativity of the square.

Now, to show the universal property, given any $P \in \mathbf{R}\text{-Mod}$ with module homomorphisms $p_M : P \rightarrow M$ and $p_N : P \rightarrow N$ such that $\mu \circ p_M = \nu \circ p_N$. If there exists module homomorphism $h : P \rightarrow M \times_Z N \subseteq M \times N$ such that $\pi_M \circ h = p_M$ and $\pi_N \circ h = p_N$, by the universality of product $M \times N$, such map h is unique if it exists. Hence, it suffices to show the existence.

For existence, the universality of the product $M \times N$ gives a module homomorphism $h : P \rightarrow M \times N$ such that $p_M = \pi_M \circ h$ and $p_N = \pi_N \circ h$. Now, for any $p \in P$, since $\mu \circ p_M = \nu \circ p_N$, one has the following:

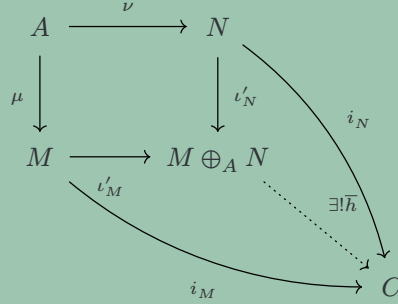
$$\mu \circ \pi_M(h(p)) = \nu \circ \pi_N(h(p)) \implies \alpha(h(p)) = (\mu \circ \pi_M - \nu \circ \pi_N)(h(p)) = 0 \quad (3.2)$$

hence, $h(p) \in \ker(\alpha) = M \times_Z N$ set wise. Therefore, the map $h : P \rightarrow M \times_Z N$ is well-defined (by restricting the codomain), which proves the desired existence.

4 6.11

Problem 4

Given any $M, N, A \in \mathbf{R}\text{-Mod}$ together with module homomorphisms $\mu : A \rightarrow M$ and $\nu : A \rightarrow N$, show that *fiber coproduct* exists in $\mathbf{R}\text{-Mod}$, i.e. an R -module $M \oplus_A N$ together with $\iota_M : M \rightarrow M \oplus_A N$ and $\iota_N : N \rightarrow M \oplus_A N$, such that the following universal property is achieved in the diagram:



Solution: First, take the coproduct / direct sum module $M \oplus N$ with canonical inclusions $\iota_M : M \rightarrow M \oplus N$ and $\iota_N : N \rightarrow M \oplus N$. Then, the compositions $\iota_M \circ \mu, \iota_N \circ \nu : A \rightarrow M \oplus N$ are two module homomorphisms that generates $\beta := \iota_M \circ \mu - \iota_N \circ \nu$.

Now, define $M \oplus_A N := \text{coker}(\beta) = (M \oplus N) / \text{im}(\beta)$, and let $\iota'_M : M \rightarrow M \oplus_A N$ and $\iota'_N : N \rightarrow M \oplus_A N$ be the compositions of the original inclusions with the canonical projection $p : M \oplus N \rightarrow M \oplus_A N$. Then, for any $a \in A$, one has the following:

$$0 = p \circ \beta(a) = (p \circ \iota_M) \circ \mu(a) - (p \circ \iota_N) \circ \nu(a) = \iota'_M \circ \mu(a) - \iota'_N \circ \nu(a) \quad (4.1)$$

This shows that $\iota'_M \circ \mu(a) = \iota'_N \circ \nu(a)$ for all $a \in A$, showing the commutativity of the given square.

Now, for the universality, given $C \in \mathbf{R}\text{-Mod}$ with module homomorphism $i_M : M \rightarrow C$ and $i_N : N \rightarrow C$ such that $i_M \circ \mu = i_N \circ \nu$, first by the universality of the coproduct $M \oplus N$, there exists a unique module homomorphism $h : M \oplus N \rightarrow C$ such that $h \circ \iota_M = i_M$ and $h \circ \iota_N = i_N$. Then, given any $a \in A$, one has the following:

$$0 = i_M \circ \mu(a) - i_N \circ \nu(a) = h \circ (\iota_M \circ \mu(a) - \iota_N \circ \nu(a)) = h \circ \beta(a) \quad (4.2)$$

Hence, this demonstrates $\text{im}(\beta) \subseteq \ker(h)$, meaning h uniquely factors through the cokernel of β , $M \oplus_A N = \text{coker}(\beta)$. Hence, there exists unique $\bar{h} : M \oplus_A N \rightarrow C$, such that together with projection $p : M \oplus N \rightarrow M \oplus_A N$, we have $h = \bar{h} \circ p$.

Therefore, one has $i_M = h \circ \iota_M = \bar{h} \circ p \circ \iota_M = \bar{h} \circ \iota'_M$, and similar logic states $i_N = \bar{h} \circ \iota'_N$.

Finally, to show the uniqueness of $\bar{h}$, if $\bar{h}' : M \oplus_A N \rightarrow C$ also satisfies the property, one has $h' = \bar{h}' \circ p$ satisfies $h' \circ \iota_M = \bar{h}' \circ (p \circ \iota_M) = \bar{h}' \circ \iota'_M = i_M$ and $h' \circ \iota_N = \bar{h}' \circ \iota'_N = i_N$ using the same logic. Hence, by the universality of coproduct $M \oplus N$, it enforces $h' = h$, hence $\bar{h} \circ p = \bar{h}' \circ p$. Using the epimorphic property of p (since it's surjective), one has $\bar{h} = \bar{h}'$. This finishes the uniqueness, hence the universality.

5 6.13

Problem 5

Prove that every homomorphic image of a finitely generated module is finitely generated.

Solution: Given $M, N \in \mathbf{R}\text{-Mod}$ with M being finitely generated, say M is generated by $m_1, \dots, m_k \in M$. Then, with module homomorphism $\varphi : M \rightarrow N$, one has $\text{im}(\varphi)$ generated by $\varphi(m_1), \dots, \varphi(m_k)$:

Given any $n \in \text{im}(\varphi)$, there exists $m \in M$, such that $\varphi(m) = n$. Yet, by finitely generated claim of M , there exists $r_1, \dots, r_k \in R$, such that $\sum_{i=1}^k r_i m_i = m$. So, $n = \varphi(m) = \sum_{i=1}^k r_i \varphi(m_i)$, showing $\text{im}(\varphi)$, can be generated by $\varphi(m_i)$, $i = 1, \dots, k$.

6 6.14

Problem 6

Prove that the ideal $(x_1, x_2, \dots)$ of the ring $R = \mathbb{Z}[x_1, x_2, \dots]$ is not finitely generated (as an ideal, i.e. as an R -module).

Solution: Suppose the contrary that the given ideal is finitely generated, then there exists $f_1, \dots, f_n \in R$, such that $(f_1, \dots, f_n) = (x_1, x_2, \dots)$.

First, by definition each of these polynomials each have only finitely many indeterminates involved, WLOG say $f_1, \dots, f_n$ only have $x_1, \dots, x_N$ involved. Then, it's clear that $(x_1, x_2, \dots) = (f_1, \dots, f_n) \subseteq (x_1, \dots, x_N)$.

However, if looking at x_{N+1} , there exists $g_1, \dots, g_N \in R$, such that $x_{N+1} = g_1 \cdot x_1 + \dots + g_N \cdot x_N$. Let's say with the involvement of $g_1, \dots, g_N$, the list of indeterminates it includes expand to $\{x_1, \dots, x_N, x_{N+1}, \dots, x_M\}$ for some $M > N$. Then, consider the following evaluation map $\varphi : \mathbb{Z}[x_1, \dots, x_M] \rightarrow \mathbb{Z}$ by $\varphi(x_1) = \dots = \varphi(x_N) = \varphi(x_{N+2}) = \dots = \varphi(x_M) = 0$, while $\varphi(x_{N+1}) = 1$.

Notice that $x_{N+1}, g_1, \dots, g_N, x_1, \dots, x_N \in \mathbb{Z}[x_1, \dots, x_M]$ based on construction, then one has $\varphi(x_{N+1}) = 1$, while $\varphi(g_1 x_1 + \dots + g_N x_N) = 0$. Yet, this is a contradiction based on the assumption $x_{N+1} = g_1 x_1 + \dots + g_N x_N$. Hence, our assumption is wrong, $(x_1, x_2, \dots)$ can't be finitely generated module over R .

7 6.15 (ND)

Problem 7

Let R be a commutative ring. Prove that a commutative R -algebra S is finitely generated as an algebra over R iff it's finitely generated as a commutative algebra over R .

8 6.16

Problem 8

Let R be a ring. A (left) R -module M is *cyclic* if $M = \langle m \rangle$ for some $m \in M$. Prove that simple modules are cyclic. Prove that an R -module M is cyclic iff $M \cong R/I$ for some (left) ideal I . Prove that every quotient of a cyclic module is cyclic.

Solution:

1. Given M a simple module (which doesn't have nontrivial proper submodule). Pick nonzero element $m \in M$, and consider the submodule $Rm \subseteq M$. It's clear that $Rm \neq 0$, so by simplicity of M , one must have $Rm = M$, showing $M = \langle m \rangle$.

2. $\Rightarrow$:

Suppose M is cyclic, $M = \langle m \rangle$. Define a map $\varphi : R \rightarrow M$ by $\varphi(r) = rm$ for all $r \in R$. This is clearly a module homomorphism (since $\varphi(sr_1 + tr_2) = (sr_1 + tr_2)m = s(r_1m) + t(r_2m) = s \cdot \varphi(r_1) + t \cdot \varphi(r_2)$). And, since $M = \langle m \rangle$, one has φ being surjective. Hence, there exists some submodule $I \subseteq R$, such that $M \cong R/I$. As a result, I is that (left) ideal.

$\Leftarrow$:

Suppose $M \cong R/I$ for some (left) ideal I . Then, take the isomorphism map $\varphi' : R/I \xrightarrow{\sim} M$, compose it with the „ R -Mod“ canonical projection $p : R \twoheadrightarrow R/I$, one has $\varphi := \varphi' \circ p : R \twoheadrightarrow M$ be a surjective module homomorphism.

Which, let $m := \varphi(1) \in M$. Since φ is surjective, all $m' \in M$ has some $r \in R$, such that $\varphi(r) = m'$. Hence, $m' = \varphi(r) = r \cdot \varphi(1) = rm$, showing that $m' \in \langle m \rangle$. So, $M = \langle m \rangle$, showing it's cyclic.

3. Given $M = \langle m \rangle$, and let $N \subseteq M$ be a submodule. Then, the projection map $p : M \twoheadrightarrow M/N$ is surjective, and it's clearly generated by $\bar{m} \in M/N$ by the surjectivity (all $\bar{m}' \in M/N$ has some $m' \in M$ as its preimage, since $m' = rm$ for some r , one has $\bar{m}' = p(m') = p(rm) = r \cdot \bar{m}$, showing $M/N = \langle \bar{m} \rangle$).

9 6.17

Problem 9

Let M be a cyclic R -module, so that $M \cong R/I$ for a (left) ideal I , and let N be another R -module.

- Prove that $\text{Hom}_{R\text{-Mod}}(M, N) \cong \{n \in N \mid \forall r \in I, an = 0\}$.
- Special case: For all $a, b \in \mathbb{Z}$, prove that $\text{Hom}_{\text{Ab}}(\mathbb{Z}/a\mathbb{Z}, \mathbb{Z}/b\mathbb{Z}) \cong \mathbb{Z}/\text{gcd}(a, b)\mathbb{Z}$.

Solution: Since $M \cong R/I$, this is equivalent to say $\text{Hom}_{R\text{-Mod}}(R/I, N) \cong \{n \in N \mid \forall a \in I, an = 0\}$.

Define a map $\varphi : \text{Hom}_{R\text{-Mod}}(R/I, N) \rightarrow N$ by $\varphi(f) = f(\bar{1})$. This is clearly a module homomorphism, since $\varphi(r \cdot f + s \cdot g) = (r \cdot f + s \cdot g)(\bar{1}) = r \cdot f(\bar{1}) + s \cdot g(\bar{1}) = r \cdot \varphi(f) + s \cdot \varphi(g)$.

First, it's injective, since if $\varphi(f) = f(\bar{1}) = 0$, all $r \in R$ has $f(\bar{r}) = f(r \cdot \bar{1}) = r \cdot f(\bar{1}) = 0$, showing $f = 0$.

Then, any $f \in \text{Hom}_{R\text{-Mod}}(R/I, N)$ has the following: For all $a \in I$, $a \cdot \varphi(f) = a \cdot f(\bar{1}) = f(a \cdot \bar{1}) = f(\bar{a}) = f(\bar{0}) = 0$, so $\text{im}(\varphi) \subseteq \{n \in N \mid \forall a \in I, an = 0\}$.

Finally, to show the image is the same as the desired set, given any $n \in N$ such that $an = 0$ for all $a \in I$, define $f : R \rightarrow N$ by $f(r) := r \cdot n$. Since for all $a \in I$ one has $f(a) = an = 0$, one has $a \in$

$\ker(f)$, or $I \subseteq \ker(f)$. Hence, this module homomorphism uniquely factors through R/I , generating a module homomorphism $\bar{f}: R/I \rightarrow N$ such that $\bar{f}(\bar{r}) = f(r) = r \cdot n$. As a result, $\varphi(\bar{f}) = \bar{f}(\bar{1}) = 1 \cdot n = n$, showing that $n \in \text{im}(\varphi)$.

Hence, $\text{im}(\varphi) = \{n \in N \mid \forall a \in I, an = 0\}$. And, with the injectivity of φ , $\text{Hom}_{R\text{-Mod}}(R/I, N) \cong \{n \in N \mid \forall a \in I, an = 0\}$.

10 6.18

Problem 10

Let M be an R -module, and let N be submodule of M . Prove that if N and M/N are both finitely generated, then M is finitely generated.

Solution: Let $n_1, \dots, n_k \in N$ generates N , $m_1, \dots, m_l \in M$ such that their image in M/N generates M/N . Then, the list of n_i 's and m_j 's in fact generate M :

Given any $m \in M$, its image $\bar{m} \in M/N$ can be written as $\bar{m} = \sum_{j=1}^l r_j \bar{m}_j = \overline{\sum_{j=1}^l r_j m_j}$, showing that $m - \sum_{j=1}^l r_j m_j \in N$. As a result, there exists $s_1, \dots, s_k \in R$, such that $m - \sum_{j=1}^l r_j m_j = \sum_{j=1}^k s_j n_j$. So, $m = \sum_{j=1}^k s_j n_j + \sum_{j=1}^l r_j m_j$, showing that M can be generated by the n_i 's and m_j 's.